LTI System Models for Random Signals

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Key Concepts

1) Random time series are often modeled as originating from a linear, time-invariant (LTI) system with white noise input.

    a) If the LTI system model is an all-pole system, the model is termed autoregressive (AR).

    b) If the LTI system model is an all-zero system, the model is termed moving average (MA).

    c) If the LTI system model has both poles and zeros, the model is termed autoregressive moving average (ARMA).

2) The power spectrum for a LTI system time series model is the magnitude squared of the system frequency response times the white noise power

    a) AR models are best suited for describing resonant or peaky power spectra by placing poles near the unit circle.

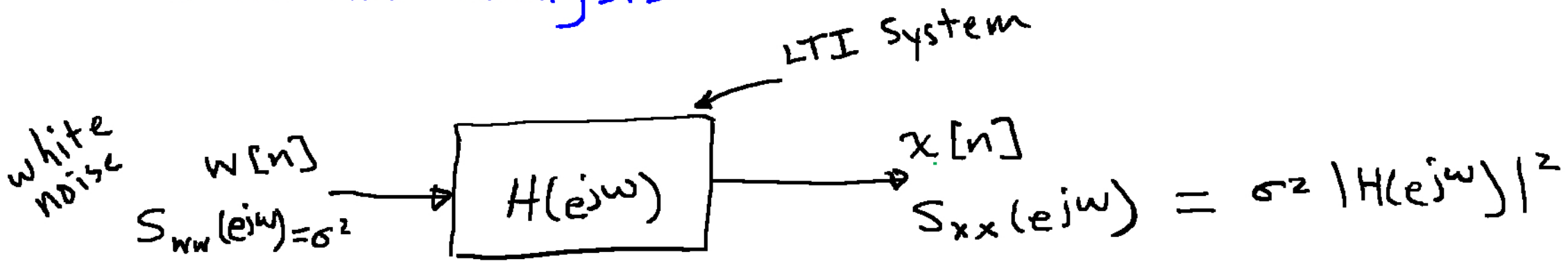
    b) MA models are best suited for describing power spectra with nulls by placing zeros near the unit circle.

    c) ARMA models have both poles and zeros and thus can describe power spectra having both resonant peaks and nulls.

3) AR models are simplest to find from data because the squared model error is a quadratic function of the unknown AR model parameters.

Video Screenshots

- Observe random signal $x[n]$. Find "model" that describes the properties of $x[n]$
- Time series analysis



Three common models

1) Autoregressive (AR): $H(e^{j\omega})$ is an all-pole system
2) Moving average (MA): $H(e^{j\omega})$ is an all-zero system
3) Autoregressive moving average (ARMA): $H(e^{j\omega})$ is a pole-zero system

Auto regressive Models (All-pole)

2:27

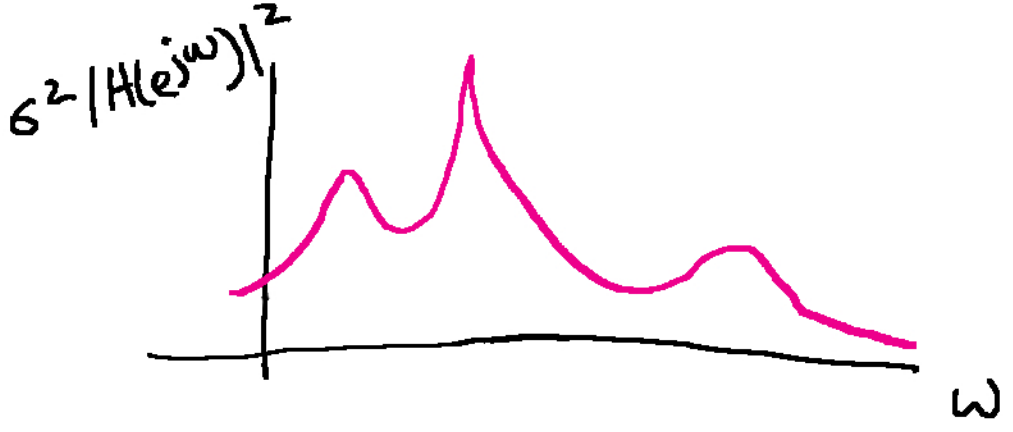
Difference equation

$$x[n] + a_1 x[n-1] + a_2 x[n-2] + \cdots + a_N x[n-N] = w[n]$$

Frequency response

$$H(e^{j\omega}) = \frac{1}{\sum_{k=0}^{N} a_k e^{-jk\omega}} \quad ; \quad a_0 = 1$$

Poles: Resonant (peaky) spectra



Moving average models (all-zero)

4:19

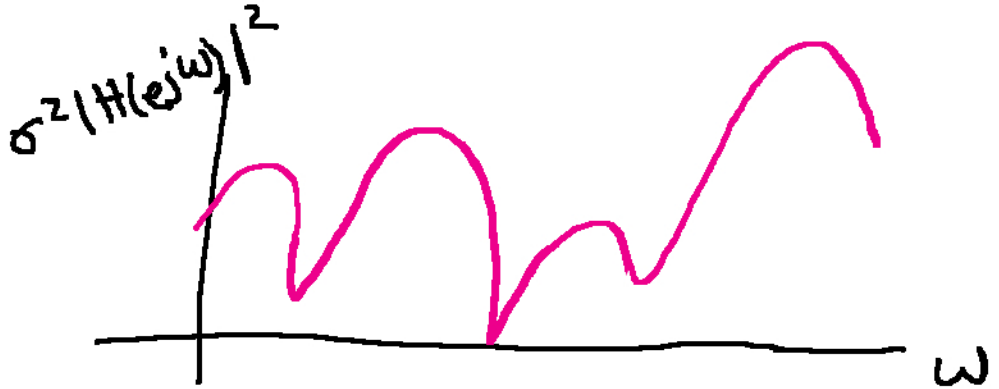
Difference equation

$$x[n] = b_0 w[n] + b_1 w[n-1] + \dots + b_M w[n-M]$$

Frequency response

$$H(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega}$$

Zeros: Nulls in spectrum



Autoregressive Moving Average (pole-zero)

5:31

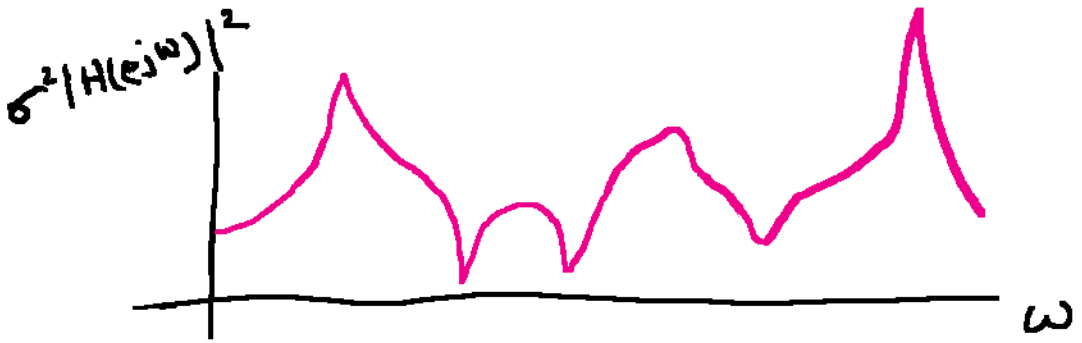
Difference equation

$$x[n] + a_1 x[n-1] + \dots + a_N x[n-N] = b_0 w[n] + b_1 w[n-1] + \dots + b_M w[n-M]$$

Frequency Response

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}} \quad ; \; a_0 = 1$$

Poles and zeros - Resonant peaks and nulls



ARMA - most general

AR - easiest to find given $x[n]$

"AR Model Error" $w[n]$

$$w[n] = x[n] - \underbrace{\left(-\sum_{k=1}^{N} a_k x[n-k]\right)}_{\text{model prediction of } x[n]}$$

$w^2[n]$ is a quadratic function of unknown a_k

$\Rightarrow$ unique, easy to find solution to minimize $w^2[n]$

"ARMA Model Error"

$$w[n] = x[n] - \underbrace{\left(-\sum_{k=1}^{N} a_k x[n-k] + \sum_{k=1}^{M} b_k w[n-k]\right)}_{\text{model prediction of } x[n]}$$

$w^2[n]$ is nonquadratic in a_k, b_k

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